TUNISIA / HEALTH INTELLIGENCE

PNSI
From public conversation to reviewed public health action

Complete research, product and engineering specification

Prepared for Nour Haouari to support discussion with ONMNE, the Ministry of Health and WHO Tunisia. This is a proposed implementation plan, not an official endorsement or an already deployed system.

WHAT THIS DOCUMENT PROVIDES
Evidence source registry and verification logic
Full page inventory, multilingual and Arabic RTL requirements
Architecture, data contracts, API boundaries and operations
Evaluation, acceptance criteria and a staged Codex build brief

11 October 2026 | Version 1.0 | Technical specification in English; French executive brief included.

Keys, institutional permissions, procurement quotations and production approvals remain to be supplied. All performance figures and scoring thresholds below are proposed targets until measured.

READING GUIDE

A specification you can build from

Attach this PDF to Codex as the master reference. Use section 33 as the starting instruction and section 36 as the acceptance checklist. The source concept note remains an input, while user decisions about accessible sources and human approval define the implementation scope.

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READING GUIDE / CONTINUED

Engineering, delivery and references

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01 / MINISTRY BRIEF

Executive brief / Résumé exécutif

La PNSI est une plateforme de veille sanitaire numérique et d’aide à la décision proposée pour l’ONMNE et le Ministère de la Santé, en lien avec les partenaires du projet. Elle observe les conversations publiques accessibles, identifie les questions et allégations sanitaires, recherche des preuves et prépare des réponses soumises à validation humaine.

Un service national, une couverture explicitement mesurée

La première version couvre des sources sélectionnées et autorisées. Elle ne prétend pas représenter toute la population tunisienne ni couvrir Facebook, Instagram ou TikTok sans accès confirmé. Les langues requises sont le français, l’anglais, l’arabe standard et le dialecte tunisien, écrit en caractères arabes ou en Arabizi. L’italien, mentionné dans la note conceptuelle pour la diaspora, reste une extension prévue.

Les capacités attendues

Collecte et recherche multilingues; regroupement des allégations; bibliothèque de preuves approuvées; analyse des préoccupations et de la propagation observée; tableaux de bord; alertes; rédaction de brouillons; validation scientifique et institutionnelle; journal des actions; suivi à 48 heures, 7 jours et 30 jours.

Une réalisation professionnelle et réaliste

Interface web dédiée avec frontend et backend séparés; prise en charge complète de l’arabe de droite à gauche; droits par institution; audit; hébergement de production en Tunisie selon le cadre validé. Les données de démonstration sont clairement signalées. Les clés API sont fournies par la porteuse du projet et ne sont jamais incluses dans le code ou ce document.

Ce dossier constitue un cahier de conception et de réalisation. Il n’atteste ni d’un déploiement opérationnel, ni de la conformité juridique, ni d’un partenariat officiellement approuvé. La validation des experts, les autorisations d’accès et les essais locaux conditionnent le passage en production.

02 / SCOPE

Scope, decisions and source of requirements

The source document is “NOTE CONCEPTUELLE (2) (1).docx”, a 17-page PNSI proposal supplied by the user. It specifies six functional layers, expert-reviewed responses, local hosting, differentiated institutional access, reporting, training, outsourced operations and a national pilot. This specification translates those ambitions into buildable requirements.

Confirmed user decisions	Implementation consequence
All health topics, not medicines only	Use an extensible taxonomy; launch with partner-selected priorities.
French, English, Arabic and Tunisian dialect	Localize the interface; process Arabic-script Derja and Latin-script Arabizi, including mixed posts.
No Streamlit product interface	Dedicated web application and API with persistent workflows.
No automatic public responses	Human-approved response versions; manual publication and recording by default.
Minimal API budget; keys supplied later	RSS/web/import baseline; optional connectors disabled until approved and configured.

Requirement language

MUST indicates a release requirement. **SHOULD** is a recommended capability that may be deferred with a documented reason. Proposed targets, scores, polling intervals and resource sizes are design assumptions, not measurements or institutional decisions.

Not in the initial scope: unrestricted access to private groups or messaging, individual diagnosis, inference of a person’s health status, automatic sanctions, claims of national prevalence, or causal attribution of health outcomes to communication alone. Broad platform coverage, Italian, large-scale audiovisual processing and epidemiological integration are gated extensions.

03 / GOVERNANCE

Users, ownership and approval rights

Role	Permitted responsibilities
Institution administrator	Manage membership, scoped roles and source configuration; cannot silently grant scientific authority.
Analyst	Review collected material, group claims, assign cases, triage and request evidence.
Scientific reviewer	Accept or correct assessments and evidence relevance; record justification.
Communication editor	Draft and adapt responses by language and channel; request review.
Institutional approver	Approve a specific reviewed response version for use by that institution.
Decision maker / partner viewer	Read authorized summaries and exports; no sensitive source access by default.
Platform operator	Operate infrastructure and jobs; no editorial approval merely by being an operator.

Small teams may combine roles under a documented policy. For high-risk claims, require an independent scientific reviewer and institutional approver. The administrator must not be able to bypass workflow gates through frontend-only controls.

Scope and accountability

Every case, assessment, response, action and report belongs to an institution or a deliberately shared workspace. Shared access is explicit and revocable. Backend authorization applies to objects, search results, aggregates, exports and attachments; a hidden menu is not a security boundary.

Assign each case an owner, reviewer, response editor, due date and escalation contact. Store transitions in an append-only audit trail. Keep ONMNE operational oversight and Ministry decision views from the note; confirm whether the proposed outsourced operating model actually applies before assigning contracts or responsibilities.

04 / WORKFLOW

End-to-end operational workflow

Stage	Input, output and mandatory gate
1. Configure	Approved source record, topic terms, access route and collection schedule.
2. Collect	Permitted public material -> provenance record; record failures and watermarks.
3. Prepare	Redact personal data, detect language spans, normalize a copy, deduplicate.
4. Understand	Identify assertions, questions, reported speech and candidate claims.
5. Group and triage	Link related mentions; analyst confirms case boundaries and priority.
6. Retrieve evidence	Search approved documents; store passages, versions and relevance.
7. Assess	AI proposes status; scientific reviewer resolves or marks uncertainty.
8. Draft and approve	Evidence-grounded draft -> language review -> institutional approval.
9. Record action	Responsible person publishes externally; record channel, time and exact version.
10. Follow up	Observe changes at 48h, 7d, 30d; document limits and lessons.

A case may be closed as a question, legitimate criticism, duplicate or irrelevant content without a corrective response. A high-harm, unverified claim can warrant urgent investigation before enough evidence exists for a conclusion. Priority and evidence status are independent.

Each stage must expose its status, last successful operation, owner and failure reason. Retries resume from the last safe checkpoint. Evidence updates may reopen an assessment; an edited response loses approval. Publication remains a human action in the initial release.

05 / RESEARCH

Research plan and feasibility gates

Work package	Concrete research output
Source audit	Named Tunisian sources, permitted routes, quotas, retention/deletion rules, language mix and usable sample yield.
Evidence audit	Partner-approved seed registry; active guidelines, local alerts, versions and document-level permissions.
Language research	Locally annotated Derja/Arabizi/code-switch test set; error analysis by script, topic and language.
AI comparison	Benchmark candidate models for retrieval, claim extraction, grounding, latency and cost.
User research	Observe analyst and approver tasks; validate case page, alert and response workflows.
Operational research	Measure ingestion uptime, reviewer workload, response turnaround and coverage changes.

First four weeks: proposed discovery sequence

Week 1: workshops with health reviewers and operators; approve taxonomy, responsibilities and data handling. Week 2: test 5-10 permitted websites or feeds plus one candidate social connector; measure actual Tunisian health-content yield. Week 3: build the seed evidence corpus and annotation guide; double-review a first claim sample. Week 4: demonstrate one complete case workflow and decide what sources and models qualify for the pilot.

Feasibility gates: a working permitted data route; accessible and current evidence; qualified reviewers; an acceptable local hosting arrangement; measurable language quality. If a gate fails, narrow the scope or use authorized manual imports. Never replace an unavailable API with undeclared circumvention or claim live coverage without a successful collection test.

06 / DATA ACCESS

Data sources: realistic access matrix

Source	Current evidence / proposed treatment
Tunisian websites and RSS	Baseline: selected feeds and permitted page retrieval. Public readability is not blanket reuse permission. Confirm terms per site.
YouTube	Official Data API with quotas [R3]. Pilot only with user key and permitted fields. Comments may be unavailable.
Bluesky	Public unauthenticated endpoints exist [R4]. Measure Tunisian relevance; search coverage is limited to the service.
Mastodon	Public timeline API documented [R5]; instance policy/authentication varies. Optional and instance-specific.
X	Pay-per-use [R6]; currently $0.005/post read. Disable by default on a zero-budget deployment.
Reddit	Explicit approval; research via RFR [R7]. Treat as pending access, not a free baseline.
Facebook / Instagram	Meta Content Library access depends on eligibility/approval [R8-R9]; free SRE route does not imply export or integration rights.
TikTok / public Telegram	Candidate extensions only. Access, jurisdiction, eligible accounts, media rights and terms not verified for this project.
Manual import	Authorized redacted text/CSV/JSON supplied by analysts; provenance and import labeling required.

Rules checked on 11 October 2026. Recheck before implementation and procurement. Source enablement requires a recorded access basis, allowed use and deletion process. News articles are monitored content; they are not automatically clinical evidence.

Measure coverage by connector, source list, topic query, period and language. Do not infer Tunisia residence from language alone. Store a Tunisia-relevance rationale such as explicit place reference or verified local publisher; expose unknown geography.

07 / INGESTION

Source registry and connector contracts

Maintain separate **monitoring_sources** and **evidence_sources**. A publisher may appear in both only after two independent approvals with different scopes. Each monitoring record MUST contain ID, institution, name, exact host/feed, platform, permitted method, permission reference, allowed fields, languages, topics, schedule, quota, retention policy, owner, last check and enabled state.

Connector interface

Implement **validate_config()**, **healthcheck()**, **fetch_batch(cursor, since, limit)**, **normalize(item)** and **apply_deletions()**. Return records, next cursor, watermark, rate-limit information and partial errors. Required item fields: source ID, platform item ID or permitted hash, published/fetched UTC timestamps, text, language hints, available engagement counts, content type, access status and provenance visibility.

Use idempotency on source plus item identity; update edits as versions instead of new independent posts. Retry transient errors with exponential backoff and jitter; respect Retry-After. Quarantine repeated schema failures and route dead-letter jobs to an operator. A missing key yields “not configured”; a denied request yields “access denied”; neither yields invented data.

Collection policy

Proposed pilot: RSS every 30 minutes; permitted website lists every 2-6 hours; API polling according to quotas. Values are configurable and measured in the pilot. Sanitize HTML and retain relevant headings. No access to private groups. Author-submitted exports require rights and redaction checks. Preserve original and normalized text separately; apply source deletion rules to cached text, vectors and derived examples.

08 / EVIDENCE SOURCES

Where the evidence whitelist comes from

Registry seed	Purpose and verification status
ONMNE - www.onmne.tn [R2]	Opened and verified as an ONMNE site with publications and thematic dossiers. Select individual official documents; do not trust the entire domain blindly.
Ministry of Health	Request the current official publication channels and approved documents directly from partners. Candidate host santetunisie.rns.tn was not readable in this check.
DPM - dpm.tn [R10]	Official role/mission page surfaced; medicine alerts and regulatory documents are candidates. Confirm current issuing authority and document validity with partners.
INEAS - ineas.tn	Candidate Tunisian clinical guideline source. Homepage retrieval failed during this research; obtain current guides from the institution.
INSP	Request current authoritative publication location and approved material from partners; no verified endpoint supplied here.
WHO guidelines [R11]	Guidelines catalog and topic-specific recommendations; check edition, population and local applicability.
PubMed / NCBI [R12]	Discovery API, not a truth whitelist. Individually review studies, reviews, retractions and accessible text.

Build a curated corpus, not an unrestricted domain filter

Start with 30-50 partner-selected documents across 3-5 pilot themes; this is a planning target. Ask reviewers for current circulars, public FAQs, national recommendations and outbreak notices. Register exact document URLs, issuing body, topic, edition, effective date, language, permitted storage and responsible reviewer. Keep documents pending until reviewed.

Additional international authorities or specialty societies may be added per topic after scientific approval. Avoid adding all journals, news websites, preprints or every PubMed record to the trusted corpus. The whitelist controls eligible evidence, but only relevant passages support a particular claim.

09 / EVIDENCE PIPELINE

Evidence governance and document ingestion

Evidence source lifecycle: **proposed -> reviewed -> approved -> suspended/retired**. A source steward proposes changes; a scientific reviewer approves scope and authoritative document types. Every change records actor, date, reason and version. Emergency source additions still require documented review.

Record	Required fields
Evidence source	Organization, exact host/path scopes, topic remit, jurisdiction, allowed document types, approval state, permissions, reviewer, review date.
Document	Title, canonical URL, issuing body, language, publication/effective/review date, version, checksum, retrieval time, permission, withdrawn/superseded flags.
Passage	Document/version ID, page/section/offset, text, extraction method and confidence, topic, embedding version.
Claim-evidence link	Supporting/contradicting/contextual relation, relevance explanation, population fit, date fit, reviewer status.

Processing pipeline

Fetch approved resources or accept authorized uploads; detect MIME type; scan uploads; extract text, with OCR only when required; preserve pages and headings; redact any accidental personal data; create semantically coherent passages; index keyword and multilingual-vector representations; validate sampled extraction against the original. The reader must be able to open the relevant page or passage.

Use provisional 400-700-token chunks with modest overlap; evaluate retrieval before settling sizes. Poll urgent alerts more often than stable guidelines. Track supersession explicitly. A document without a date is not automatically current. If a critical source is unavailable, display the failure and allow authorized manual upload; never manufacture a quote or cite an unseen full text.

10 / VERIFICATION

Claim extraction, verification and abstention

Identify atomic, checkable claims while preserving context, negation, timing, population, quantity and source stance. Separate author assertions, questions, sarcasm, quotations and reported rumors. A post can contain several claims. Merge only equivalent claims; “can cause” and “always causes” must not be treated as the same statement.

Verification sequence

Create a neutral claim representation in the original language and a retrieval translation. Search the approved local corpus first, then approved international sources and NCBI discovery where appropriate. Retrieve both supporting and contradicting material. Rerank by meaning, authority, currentness and local applicability. Produce a structured assessment linking each conclusion to passages. A scientific reviewer confirms or changes it.

Evidence status	Meaning
Unassessed	No substantive verification completed.
Supported	Evidence supports the claim in the stated context.
Contradicted	Relevant evidence contradicts it.
Misleading / missing context	Some factual content but an unjustified interpretation, omission or generalization.
Insufficient evidence	Not enough reliable evidence to decide.
Conflicting evidence	Credible relevant sources disagree; specialist resolution needed.
Not a factual claim	Question, opinion or criticism requiring a different response.

“Rumor” describes circulation of an unverified claim, not a false verdict. “Disinformation” implies intent; do not infer intent from text classification alone. Critical disagreement with institutions is not evidence of falsehood. Keep machine proposals distinct from reviewer-confirmed assessments and provide a correction/reopening mechanism.

11 / ANALYTICS

Scores, indicators and alert thresholds

Maintain three separate values: **priority** for action, **evidence status** for the claim, and **model confidence** only when calibrated. Do not display an invented “truth percentage”. A model’s self-reported certainty is not a measured probability.

Proposed triage score, subject to validation

Priority = 100 × (0.45H + 0.25G + 0.15E + 0.15V), where H is expert-assessed potential health harm, G is growth within observed source coverage, E is available engagement normalized within platform, and V is exposure proxy based on collected mentions. Components are scaled 0-1, with explanations. These weights are initial design choices, not WHO-endorsed values.

When data are missing, show score completeness and a provisional/manual priority rather than silently treating unknowns as zero. Do not compare like counts directly across platforms. Use stable-source baseline windows and mark API outages or source additions. Never call a collected-post count “people reached” without a valid reach measure.

Proposed alert rule	Guard / action
Expert flags immediate potential harm	Escalate even with low volume; evidence may remain uncertain.
At least 10 deduplicated mentions and >=3x baseline growth	Analyst review; a coverage change must not be mistaken for a surge.
Priority >=70; >=40 medium	Pilot thresholds only; tune against reviewed cases and capacity.
Repeated signal for same case	Deduplicate, apply cooldown; reopen for meaningful changes.

The concept note’s “infodemic temperature” should remain an experimental composite with visible components, denominator and uncertainty. Never present it as an established national metric. Alerts need an owner, rationale, due time, acknowledgement and escalation history.

12 / LOCALIZATION

Languages, Derja, Arabizi and accessibility

Layer	Mandatory behavior
UI locales	French default; English, standard Arabic and reviewed Tunisian Derja interface strings. Language selection persists per user.
Content handling	Arabic-script Derja, Arabizi, French, English, standard Arabic and mixed-language spans. Unknown language is allowed.
Response variants	French, English, Arabic and Derja, with separate Arabic-script and Arabizi variants where requested; approve each variant.
RTL layout	Arabic/Arabic-script Derja mirror layout; Latin Arabizi remains LTR. Use bidirectional isolation for IDs, URLs and mixed text.
Future extension	Italian from the concept note; prepared locale slot, disabled until translations and evaluation are ready.

Keep original text unchanged; store normalization and translation as derived views. Arabizi digits can represent sounds, but mappings are contextual: do not apply a universal destructive digit replacement. Retain negation, emojis, punctuation and clinical units when they alter meaning. Mark uncertain transcription and translation; allow reviewer correction.

Search and rendering

Use language-aware keyword expansion plus evaluated multilingual embeddings. Query in one language should retrieve equivalent claims in another; expose original and translation side by side. RTL tables, forms, charts, drawers, pagination and PDFs require testing. Dates and number formats follow locale; timestamps remain UTC internally and display with an explicit timezone.

Accessibility target: WCAG 2.2 AA design practices, verified in testing. Keyboard navigation, visible focus, chart table alternatives, readable contrast, scalable text, descriptive labels and status words alongside colors are required. Derja medical wording and translated corrections must be reviewed locally; translation alone does not establish scientific equivalence.

13 / ENGINEERING

Architecture and technology choices

Recommended baseline: a **modular backend** with asynchronous workers, rather than many independent microservices before the pilot. This preserves clear domain boundaries without imposing Kafka, Kubernetes, Neo4j or several databases from day one. Split services only after measured scale or isolation requirements justify it.

Component	Proposed implementation
Web frontend	Next.js, React, TypeScript; accessible component system, Tailwind styling, localized routing [R13].
API	FastAPI with typed request/response models and generated OpenAPI [R14]; domain services and centralized authorization.
Persistent data	PostgreSQL for cases, roles, workflows and audit; pgvector as an optional evaluated vector extension.
Files	Locally hosted S3-compatible object storage; private signed access, malware scanning and retention jobs.
Workers	Celery + Redis, with a scheduler, persistent job records and idempotency; separate CPU/optional GPU queues.
Models	Provider adapters for local or approved external inference; separate extraction, retrieval/reranking and drafting roles.
Identity	Institutional OIDC when available; secure local accounts for isolated development only.
Operations	Containerized services, reverse proxy/TLS, health checks, metrics, logs, backups and deployment runbook.

Data path: browser -> API -> PostgreSQL/private storage; API -> queue -> workers -> permitted sources or approved model adapters; workers -> evidence/search indexes -> API. The browser never calls a secret-bearing social or model API directly.

Keep collection, evidence, inference, review, action and reporting modules separate. Adopt maintained stable dependency versions at implementation time, lock them and scan licenses/vulnerabilities. No version number here is a promise that a package will remain supported. These are engineering proposals, not a prescribed ministry stack.

14 / DATA CONTRACTS

Persistent data model and provenance

Entity	Core fields and relations
Institution / membership	UUID, institution, user, role, scope, active state; membership audit.
Source / collection run	Permission record, config reference; run timestamps, cursor, counts, failures, coverage snapshot.
Content item / revision	Source identity, UTC times, redacted original, normalized text, languages, metrics, deletion status.
Claim / mention / case	Atomic claim, context, language, linked mention IDs, owner, status, topic and priority explanation.
Narrative / assignment	Cluster version, case members, merge/split history, analyst confirmation.
Document / passage	Evidence source, version, checksum, permissions, page offsets and extraction confidence.
Assessment / evidence link	Claim, status, rationale, passage IDs, contradictions, model run, reviewer and review time.
Response / variant / approval	Exact text/version, language/script, cited assessment version, reviewers, approvals and invalidation.
Action / follow-up	Approved version, publisher, channel, time, baseline, windows, coverage and outcomes.
Alert / report / audit / job	Owner and transitions; query snapshots; immutable actor events; retry and task status.

Foreign keys enforce referential integrity. Use optimistic concurrency via version columns; all mutable records have created_at, updated_at and deletion policy. Index institution, source, date, topic, state and case relations. Deduplication must not erase provenance or inflate count of independent claims.

Every output can be traced to source revision, processing/model version, prompt version, evidence document version and reviewer action. Keep organizational accounts separate from individual identifiers. Use anonymized examples and aggregates in exports. Restricted provenance storage is a deliberate privacy decision, not a default feature.

15 / API

API contract and integration boundaries

All application endpoints live under **/api/v1**. Require authenticated scoped access except /health/live and tightly limited readiness. Publish OpenAPI and generate typed frontend clients. Cursor pagination uses limit <=100; filters include time interval, topic, source, language, evidence status and institution scope.

Endpoint family	Operations
GET /me; GET /overview	Current role/locale; aggregate dashboard with coverage and filter snapshot.
/sources; /collection-runs	Read/configure permitted sources; operator validation and enablement.
/content; /imports	Search redacted items; inspect revisions; upload validated imports.
/claims; /cases; /narratives	Create, assign, merge/split, triage; GET /cases/{id}/timeline.
/evidence-sources; /documents	Source review; ingest approved docs; inspect passages and supersession.
POST /claims/{id}/verify	202 + job ID; GET /jobs/{id} returns queued/running/succeeded/failed.
/assessments/{id}/review	Scientific decision with rationale and expected_version.
/responses; /responses/{id}/approve	Draft version, request reviews, approve exact variant/version; never automatic publication.
/actions; /follow-ups; /alerts	Record publication, schedule observations, acknowledge/escalate alerts.
/reports; /audit; /users	Queued exports, authorized audit view and scoped membership management.

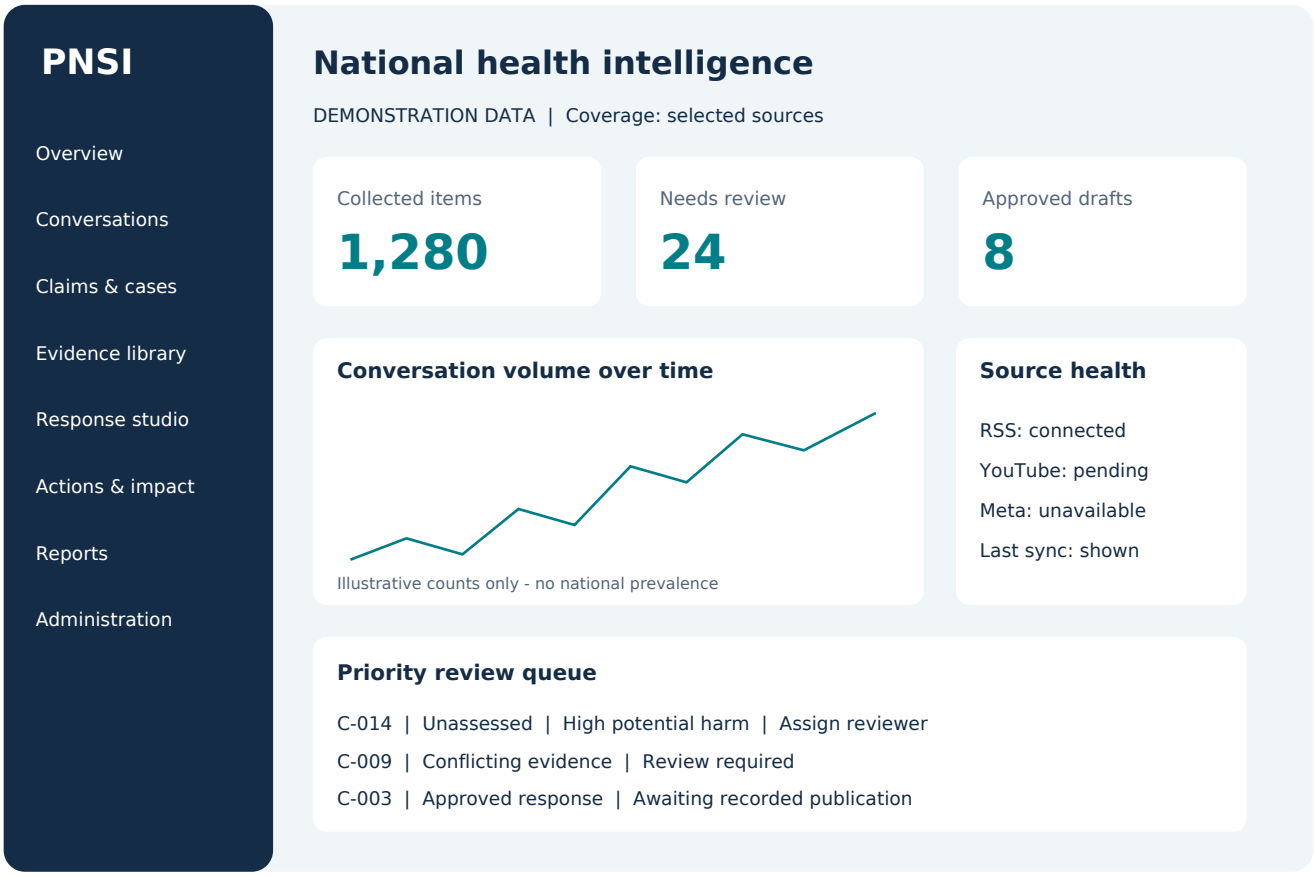
Mutation requests include expected_version and, for retry-sensitive operations, Idempotency-Key. Errors use code, safe message, field errors and request_id; never expose secrets. Return 401/403 for authorization, 409 for version conflicts, 422 for validation, 429 with retry guidance and 503 for unavailable dependencies. Authorization checks occur before counts or file access.

Background job progress uses polling first, optional server-sent events later. Exports are scoped, expiring private downloads. Avoid unsafe user-supplied fetch URLs; only validated source scopes can trigger remote retrieval.

16 / PRODUCT DESIGN

Visual direction: ministry-grade interface

Create a calm institutional workspace: white and light-grey surfaces, deep navy navigation, teal actions, restrained amber/red risk accents, strong typography and generous spacing. Professional depth comes from clear evidence, filters and workflow states. Do not add decorative 3D effects, fake national maps or invented live metrics.



This is a proposed layout, not a running platform. All numbers above are demonstration values. Use a persistent language switcher, institution selector, search, notifications and profile menu. Separate priority from evidence badges. Display source health and coverage beside statistics.

Recommended layout: left navigation, compact global filters, KPI cards, trend chart, review queue and source-health panel. At 1440 px, provide a two-column case workspace; at tablet widths stack panels; mobile supports reading and urgent review. Provide loading, empty, partial, error and restricted states for every component.

17 / PAGES

Page inventory I: monitoring and investigation

Page / route	Required content and interactions
Sign in /login	Secure account/SSO, recovery, session handling, locale selector; generic error messaging.
Overview /overview	Observed counts, priority cases, unreviewed assessments, trends, source freshness and filter-aware coverage. Drill down from every KPI.
Conversations /conversations	Search/filter, original + translation, source/date, assertion/question stance, duplicate links and redacted context. Open case or create claim.
Import /imports	File template, validation preview, provenance declaration, redaction preview, job results and row-level errors.
Claims /claims	Evidence status, reviewer state, priority, topic, source diversity, owner and last checked date; bulk assignment only.
Case workspace /cases/{id}	Claim, related posts, spread, evidence, assessment, response, action and history tabs. See next section.
Narratives /narratives	Theme clusters, growth, representative redacted examples, merge/split and analyst-confirmed labels.
Alerts /alerts	Severity, rationale, evidence status, coverage caveat, owner, due date, acknowledge and escalation.
Evidence registry /evidence/sources	Approved/pending/suspended sources, topic remit, permissions, owner, freshness and source approval workflow.
Evidence library /evidence/documents	Full-text search, versions, page/passage viewer, supersession, rights and extraction quality.
Evidence review /evidence/reviews	Proposed source/document changes, disagreements and expert decisions.

Every page MUST have functioning backend data, permission checks, pagination where relevant and shareable filter state. Empty data is explained; it is not replaced with demonstration metrics outside a visibly separate demo mode. Search results must not leak restricted institutions.

18 / CASE WORKFLOW

Case workspace: the central operational page

The case header displays ID, neutral claim wording, original language, topic, evidence status, machine-versus-reviewed label, priority/completeness, owner and last review date. The header remains visible as the reviewer moves between tabs.

Tab	Contents and actions
Summary	Why the case matters, unresolved questions, next task, harm rationale and reviewer assignments.
Conversation	Redacted mentions, translations, context, duplicates and source availability; attach/detach items with a reason.
Spread	Observed volume by source/time, baseline, sampling limits, available engagement; unknown region bucket.
Evidence	Supporting/contradicting/context passages, document dates, exact page links, fit and source approval state.
Assessment	Structured proposed verdict, uncertainty, reviewer correction, disagreement and version comparison.
Responses	Channel-specific language variants, citations, review checklist, approval state and exact version.
Actions	Publication recorded by a person, responsible institution, channel, date, approved version and permitted reference.
Follow-up	48h/7d/30d observations, coverage comparability, trend interpretation and confounders.
Timeline	All assignments, model runs, evidence changes, reviews, approvals and actions with actor and timestamp.

Primary actions follow role and state: assign reviewer, request verification, accept/revise assessment, create draft, request approval, approve exact version, record action or reopen. Disabled actions include a reason. Scientific and communication review are distinct checkpoints.

Resolve concurrency visibly: if another reviewer updates the case, present the difference and ask for reload or reconciliation; never silently overwrite. When cited evidence is withdrawn, mark affected assessments and drafts stale and require review. A published response may need a recorded correction, not deletion of its history.

19 / PAGES

Page inventory II: responses, operations and oversight

Page / route	Required content and interactions
Response studio /responses/{id}	Rich text with cited evidence, original draft, language/script variants, channel limits, translation review and version comparison.
Approval queue /approvals	Pending scientific/language/institutional checks; reviewer assignments; approve, reject or request changes with reason.
Actions /actions	Recorded communications, channels, institution, date, exact approved content and follow-up status.
Impact /impact	Comparable before/after windows, observed conversation measures, coverage changes, external indicators and caveats.
Reports /reports	Daily crisis, weekly routine, monthly strategy; preview, approved export and snapshot history.
Sources /sources	Monitoring coverage, access basis, quota, last sync, errors, enablement and permission expiry.
Annotation /annotation	Original/context, task labels, uncertainty, independent annotators, disagreement resolution and dataset version.
Model evaluation /evaluation	Metrics by language/task, frozen datasets, errors, calibration, cost/latency and release comparisons.
Administration /admin	Institutions, users, roles, taxonomy, reviewed translations, alert policies, retention and settings.
Jobs /operations/jobs	Queue health, failed tasks, retry/cancel where safe, connector errors and quota.
Audit /audit	Authorized filtered event search and export, before/after versions, request ID and actor.
Help /help; profile /profile	Role-specific SOPs, glossary, keyboard help, locale/timezone and notification preferences.

Institutional onboarding creates membership, role and topic scope, not public account sign-up by default. A notification panel links to a real alert or task. Search, save, export and approval controls must work end to end; do not ship decorative buttons. No public portal is required for the initial scope.

20 / RESPONSE WORKFLOW

Response generation and human sign-off

Input to drafting: reviewer-confirmed claim assessment, approved evidence passages, intended audience, language/script, channel and approved institutional communication guidance. If evidence is unresolved, draft an uncertainty-aware explanation or investigation notice instead of asserting a correction.

Draft structure

Lead with the accurate information in simple language; acknowledge the underlying concern; explain the relevant evidence and limits; provide a practical next step agreed by reviewers; include references and a checked date. Avoid sensational repetition of a harmful claim, stigmatization, personal medical instructions or fabricated official attribution. Channel adaptations can include posts, FAQs, spokesperson Q&A; and short video scripts.

Approval state machine

Draft -> scientific_review -> language_review -> institutional_review -> approved -> action_recorded -> follow_up -> closed. Rejection returns to changes_requested with a reason. Individual variants can have different states; approval of French does not approve Arabic or Arabizi. Only reviewed current evidence is eligible for an approved factual correction.

Approval binds response ID, immutable version, language, script, cited assessment version, evidence snapshot and approver. Editing any factual wording, translation, citations or assessment invalidates the affected approval. The backend enforces this even if a client directly calls the approve endpoint.

No automatic external publishing in the baseline. Copy/export the approved variant, have the authorized person share it, then record what actually occurred. Optional publishing integrations require explicit institutional scope, platform access and a separate evaluated implementation. AI-produced text remains clearly labeled until reviewed.

21 / EVALUATION OF ACTIONS

Actions, follow-up and impact interpretation

An action record links to the case, approved response version, institution, responsible person, channel and actual publication time. Record whether it was a post, FAQ, press statement, broadcast, local engagement or another intervention from the concept note. A planned action is not counted as completed.

Follow-up schedule

Schedule observations at 48 hours, 7 days and 30 days after the recorded action. Capture a defined baseline and post-action window, available mentions, unique collected items, available engagement, narrative mix, question volume and estimated expressed emotion. Record sampling, deleted items, source outages and changes in search terms.

Compare only compatible coverage. With a zero or very small baseline, avoid unstable percentage changes and show absolute counts. Display reported region only when reliable; do not infer personal residence or health status. The dashboard describes measured conversation changes, not population behavior.

External indicators

Partner-provided vaccination, consultation or hotline aggregates may be attached if authorized and appropriately documented. Keep provenance, period, definition and geography separate from social-media metrics. Integration requires an agreed schema and data-sharing arrangement; no national health-system API is assumed.

Before/after changes do not prove causal effects. For stronger evaluation, involve an epidemiologist or evaluation specialist and predefine comparison groups or interrupted time-series designs where suitable. Note concurrent events, campaigns and collection changes. Record useful lessons and reviewer interpretation without claiming that a generated response caused a health outcome.

22 / REPORTING

Reports, exports and decision briefs

Provide daily crisis reports, weekly monitoring summaries and monthly strategic reports, reflecting the concept note. A report contains scope/period, data coverage, new and unresolved claims, evidence status, high-priority cases, reviewed insights, approved responses, recorded actions, follow-up findings and limitations.

Reproducibility

Store query filters, source set, collection-run cutoffs, metric definitions, model and evidence versions, creator, institution and export time. A saved report is a dated snapshot; later source changes must not silently alter it. Rebuilds create a new version. Aggregation honors item deletion and retention obligations.

Formats

PDF for decision makers; CSV/XLSX where authorized for aggregates; JSON for approved machine-readable interchange. Escape spreadsheet formula injection in user-derived cells. Export templates support French, English and Arabic RTL; Derja summaries require language review. Preserve citation links and source-version references without exporting personal identifiers.

Decision brief: one-page concern summary, observed trajectory, public-health risk rationale, what is known/unknown, action options, responsible team and next review time. AI proposes text; assigned experts approve scientific and institutional content. Report recipients and notification channels are configured by the institution, not inferred from a project name.

Useful KPIs include reviewed-case turnaround, open backlog, evidence freshness, approved-response time, source availability and percentage of machine proposals corrected by reviewers. Do not equate “number of false claims detected” with system quality without measuring errors and sampling limits.

23 / DATA GOVERNANCE

Privacy, data residency and the actor-map conflict

The concept note requires no retained personal identifiers, while also requesting maps of accounts, influencers and propagation networks. These are not automatically compatible. Pseudonymized IDs and searchable text can still allow identification. Removing a username does not establish anonymity.

Baseline design choice

Use aggregate narrative/source maps and organization-level publisher information where approved. Do not build individual influencer profiles or personal health inference in the initial release. Redact names, handles, phone numbers, email addresses and sensitive identifiers from working examples; minimize original text storage and review residual identification risk.

Two provenance options require institutional resolution: (A) strict minimization, retaining source/document identity without personal post links; or (B) a restricted provenance vault for authorized traceability, with defined lawful basis, retention and access audit. Option B departs from a literal “no identifiers retained” policy and cannot be enabled merely for convenience. Use A as the default until this is resolved.

Residency and retention

Production data hosting in Tunisia is a requirement in the note. External model, OCR, logging or email services may transfer data even if the database is local. Use local processing by default for sensitive material; external processing requires a recorded institutional decision. This document does not certify legal compliance.

Proposed starting retention: raw ingestion buffers <=24h; redacted working items <=90 days; evidence according to rights/currentness; action/audit history according to institutional policy. These values are provisional. Deletions must cover storage, caches, embeddings, examples and recoverable backups according to the approved policy. WHO ethics guidance is a relevant framework [R1], not a substitute for local review.

24 / SECURITY

Security controls and misuse resistance

Boundary	Required control
Accounts	Institutional OIDC or secure approved auth; MFA for privileged roles; session expiry and revocation.
API	Object-level scoped authorization, CSRF protection where cookies are used, origin controls, rate limits and input validation.
Uploads	Size/type limits, malware scan, isolated parsing, archive-bomb protection and private storage.
Source fetching	Allowlisted hosts/path scopes, redirect validation, SSRF protection, timeouts and no private-network fetching.
Secrets	Server-side secret store/env injection; no browser bundle, repository, PDF, logs or exported credentials.
Data	TLS in transit, encryption at rest, least privilege, scoped signed downloads and audited access.
AI	Treat posts and documents as untrusted content; no execution of embedded instructions or unrestricted tools.
Operations	Dependency scanning, patch schedule, tested backup restore and incident response.

Prompt injection can arrive in a rumor, website or PDF. Retrieval content cannot change the system’s workflow, source approvals or permission checks. Limit model output to validated schemas; do not execute generated SQL or commands. Block citations that do not resolve to stored evidence IDs. Model tools may fetch only registered evidence/source scopes.

Audit sensitive access, exports, source changes, role changes, reviews and approvals. Redact logs and model traces. Review denial-of-service and job-cost limits. A breached or revoked source is disabled without erasing evidence of prior decisions. Threat modeling and privacy review are release tasks, not a promise that all risk is eliminated.

25 / AI

AI implementation and model-selection strategy

Use separate tasks with typed outputs: health-topic relevance, language spans, assertion/question stance, atomic claim extraction, semantic matching, narrative clustering, evidence retrieval/reranking, assessment proposal and grounded response drafting. Start with evaluated pretrained components; do not assume custom training is necessary for every task.

Benchmark before selection

Candidate local models must be tested on Tunisian Derja, Arabizi, Arabic, French and English. Compare against a simple keyword/lexical baseline and an approved hosted model only if policy permits. Record license, hardware needs, per-task quality, abstention, latency, inference cost and language errors. No named model is declared best without this benchmark.

RAG flow: approved corpus -> versioned passages -> multilingual keyword/vector retrieval -> reranking -> evidence-fit checks -> constrained assessment with passage IDs -> expert review. Embedding similarity finds related text; it does not verify a claim. A language model should not act as the sole source of truth.

Structured output and failure modes

Assessment output includes claim ID, proposed status, rationale, supporting/contradicting passage IDs, missing evidence, context limitations and model-run ID. Validate JSON with typed schemas and bounded retries; unresolved parse/grounding failures produce “review required”. Set generation budgets and maximum evidence context sizes.

Keep prompt templates versioned in code, with controlled release changes. Run model regression evaluations when prompts, corpus, embeddings or model weights change. Never train on collected platform data unless permission allows it. Reviewer decisions become a versioned annotation dataset only under the approved data-handling plan.

26 / VALIDATION

Evaluation datasets and scientific validation

Create a locally reviewed benchmark with original-language posts, context, atomic claims, stance, topic, evidence relevance, proposed status and response-quality judgments. Suggested pilot target: 800-1,500 annotated items and 150-300 claim-evidence cases, subject to availability and expert capacity. These are planning figures, not existing datasets.

Avoid leakage

Group near duplicates and equivalent claims before splitting. Keep narrative families and translated versions in the same partition; use a later-time holdout for emerging claims. Separate development, validation and frozen test sets. Do not tune on the final test set or feed its labels into prompts. Use independently reviewed labels and adjudicate disagreement.

Task	What to measure
Topic / stance / claim extraction	Precision, recall, macro-F1; negation, questions and quotation errors.
Retrieval	Recall@k, ranking relevance, supporting and contradicting evidence discovery.
Assessment	Per-status confusion matrix, dangerous false assurance, false accusation and abstention coverage.
Drafting	Citation validity, passage support, factual errors, omissions, tone and language equivalence.
Clustering	False merges/splits; same claim vs same topic distinction.
Operations	Turnaround, review burden, source uptime, job failure and cost per reviewed case.

Report each metric by language/script, mixed text, topic and content type, with sample size and uncertainty. Double annotation on a substantial subset helps assess agreement. Proposed targets must be approved by the scientific team; no AUROC or accuracy is claimed in this document.

Model confidence is displayed only after calibration on suitable held-out data. Maintain human review for final assessments even if benchmark results are strong. Regularly review drift caused by new slang, campaigns, platform changes and outdated guidance.

27 / TESTING

Functional tests and release evidence

Test scenario	Required result
Missing social API key	Connector is visibly not configured; remaining sources and manual import work.
Repeated import / retries	No duplicate mentions or repeated notifications; provenance retained.
Question quoting a rumor	Not treated as the author promoting a false claim; uncertainty preserved.
Conflicting / absent evidence	No forced supported/contradicted result and no definitive corrective draft.
Arabic RTL and mixed Arabizi	Readable layout, intact original meaning, usable forms and correct script direction.
Edit an approved translation	Affected approval is invalidated; action cannot cite an unapproved revision.
Wrong institution accesses case/export	Backend returns denied access; no aggregate or download leakage.
Malicious text/PDF instructions	No permissions, tool actions or evidence approval changed by content.
Source outage / quota hit	Partial coverage clearly shown; no false zero-volume or surge interpretation.
Deletion / superseded evidence	Cached content removed as required; impacted decisions marked for review.
Record actual publication	48h/7d/30d follow-up tasks created once against exact approved version.

Automate meaningful unit tests for score completeness and approval invariants, integration tests for database/queue/API boundaries, and browser end-to-end tests for role-based case handling and RTL. Test restore, secret redaction and access isolation. Run manual scientific review on representative language cases.

The release report identifies build commit, environment, dependency lockfiles, test datasets, passed/failed checks, known limits and sign-off owners. Screenshots alone do not prove functionality; a complete persisted workflow and real permitted connector test are required for a live pilot.

28 / RESOURCING

Resources, staffing and realistic costs

Resource	Planning assumption / responsibility
Application pilot host	Provisional 4-8 vCPU, 16-32 GB RAM, 100-250 GB encrypted SSD; CPU workloads and modest corpus only. Benchmark before buying.
Local model inference	Separate machine or approved shared capacity; GPU need depends on model size, quantization, context and concurrency. No hardware guarantee.
Low-resource development	Laptop + containers + small corpus + manual imports; optional models/ASR can be disabled.
People	Product/technical lead, web/backend engineer, AI/data engineer, scientific reviewer, communication/language reviewer and operator. Roles may be combined.
Institution inputs	Current evidence, source approvals, taxonomy, reviewer availability, hosting/security policy and communication guidance.
Operating costs	Hosting, storage/backups, inference, paid data access, maintenance, annotation and expert review. Software being open source does not remove these costs.

Use a quotation worksheet: monthly fixed hosting + backups + security/monitoring + API reads × current rate + model tokens or compute + staff effort + contingency. Obtain Tunisian hosting and service quotations; no unverified TND prices are invented here. At the current X read price, 10,000 returned posts cost about USD 50 for those reads alone [R6].

Start text-first; process audiovisual material only when permitted and prioritized. Cache reusable evidence, deduplicate mentions and use smaller models for easy routing. Set daily/monthly job and API budgets. Funding is an institutional decision; the platform must remain usable with manual import and a local evidence corpus when paid integrations are disabled.

29 / OPERATIONS

Deployment, reliability and support

Development uses Docker Compose with frontend, API, worker, scheduler, PostgreSQL, Redis and private object storage. Provide a one-command documented startup, migrations, a safe synthetic seed and health checks. Use separate development, staging and production configurations and credentials.

Production readiness

Choose approved Tunisian hosting, TLS/domain, identity provider, storage, logging and backup destinations. Document any external egress and inference. Build signed/versioned images in CI; run tests and scans; deploy to staging; perform reviewer UAT; deploy the exact approved build with rollback. Kubernetes is optional, not a prerequisite.

Proposed service targets

For the pilot, aim at 99% monthly availability for the web workspace, p95 <2 seconds for common read pages under the measured pilot load, and asynchronous progress for expensive verification. These are proposed goals, not contractual SLAs. Define source freshness separately; API collection latency cannot exceed vendor permissions or quotas.

Set provisional recovery objectives of RPO <=24h and RTO <=8h, then validate with the hosting team. Perform encrypted backups, integrity checks and a documented restore drill before pilot launch. Record uptime, queue depth, retry rates, database growth, source failures, evidence staleness and cost consumption.

Runbooks cover revoked credentials, API change, unavailable evidence, broken ingestion, incorrect approved response, suspected breach, backup recovery and rollback. Assign business-hours/on-call coverage explicitly. The note’s continuous monitoring ambition cannot imply 24/7 human response unless staffing and funding support it.

30 / DELIVERY

Delivery roadmap and gated milestones

Milestone	Exit evidence
M0: discovery and governance	Approved source/evidence registry plan, privacy choice, taxonomy, roles, acceptance criteria and hosting constraints.
M1: product foundation	Localized dedicated interface; authentication, database, scoped roles and persistent case workflow.
M2: first complete vertical slice	Permitted import/RSS -> claim -> evidence -> expert assessment -> draft -> approval -> recorded action.
M3: monitored pilot	Approved live sources, source-health view, alerts, multilingual evaluation and trained reviewers.
M4: operational stabilization	Reliable reports, 48h/7d/30d follow-up, restore drill, audit and measured workflow quality.
M5: selective expansion	Only qualified APIs, audiovisual features, Italian and epidemiological integration after separate gates.

Proposed engineering sequence: weeks 1-4 discovery; weeks 5-8 foundation and vertical slice; weeks 9-12 retrieval/review/drafting; weeks 13-16 pilot, reporting and hardening. This assumes a staffed team and timely inputs; it is not a promise for one student to deliver a national system in four months.

The concept note labels a 12-month pilot, but its phase durations total 10-14 months if sequential. Resolve this by overlap or a revised schedule. A feasible national service plan may span 12 months: months 1-2 cadrage; 3-5 core build; 6-9 operational pilot; 10-12 consolidation. Institutional decisions and access approvals may shift dates.

Every milestone produces a reviewable build, documentation, tested workflows and known limitations. A polished ministry demo can use synthetic data, but must label it and remain distinct from the live pilot. Do not advertise unavailable connectors as delivered.

31 / DECISIONS

Unresolved decisions and risk register

Decision / risk	Proposed treatment and owner
Access to dominant social platforms	Partner/source steward confirms eligibility and permissions; baseline remains narrower and transparent.
No identifiers vs actor mapping	Privacy lead chooses strict aggregate baseline or approved restricted provenance; individual maps deferred.
Local hosting vs hosted AI	Security/institutional lead approves processing locations and egress before sending data.
Current local clinical/regulatory sources	Scientific lead provides current documents and authentic publication channels.
Derja UI and clinical terminology	Local language reviewer approves terminology, script choice and translations.
Scoring and thresholds	Scientific/operational leads validate against annotated cases and reviewer capacity.
Staffing and outsourcing	ONMNE confirms ownership, response coverage and whether the provider model applies.
Automatic publishing	Not required; manual publication baseline. Separate approval/integration if later requested.
Representativeness and causality	Show coverage limits; involve evaluation specialists for population or impact claims.
Institutional branding	Use neutral proposal styling; logos and endorsement wording only with authorization.

Maintain a living decision log with ID, issue, owner, due date, chosen option, evidence and affected requirement IDs. Codex may implement reversible technical defaults, but must not invent source permissions, scientific verdicts, institutional authority or API keys.

Critical risk controls are incorporated into the product: incomplete coverage warnings, evidence abstention, source freshness, human approval, original-language views, auditability and budget caps. Review them as operational behavior rather than burying them in a disclaimer.

32 / CONFIGURATION

Configuration and API-key placeholders

Provide **.env.example** containing names and safe placeholders only. Actual values are supplied by Nour or the institution through local ignored files or a deployment secret manager. Every provider adapter reads server-side configuration. Missing optional credentials disable that connector without breaking the core workflow.

Variable group	Suggested names / notes
Core	APP_ENV, APP_BASE_URL, DEFAULT_LOCALE=fr, DEFAULT_TIMEZONE=Africa/Tunis, DATABASE_URL, REDIS_URL
Identity	OIDC_ISSUER, OIDC_CLIENT_ID, OIDC_CLIENT_SECRET; development auth guarded by environment.
Storage	S3_ENDPOINT, S3_BUCKET, S3_ACCESS_KEY, S3_SECRET_KEY; locally hosted production endpoint.
AI	AI_PROVIDER=local|approved_remote, AI_BASE_URL, AI_API_KEY, EXTRACTION_MODEL, DRAFT_MODEL, EMBEDDING_MODEL, RERANKER_MODEL
Evidence discovery	NCBI_EMAIL, NCBI_TOOL, NCBI_API_KEY optional; SEARCH_PROVIDER and SEARCH_API_KEY only if chosen/approved.
Optional collection	YOUTUBE_API_KEY, X_BEARER_TOKEN, REDDIT_CLIENT_ID/SECRET; approval is required separately from credentials.
Public/instance services	BLUESKY_BASE_URL, MASTODON_INSTANCE_URL/TOKEN; access policy still applies.
Governance	EXTERNAL_AI_ENABLED=false, AUTO_PUBLISH_ENABLED=false, RAW_RETENTION_HOURS, REDACTED_RETENTION_DAYS, DAILY_BUDGET
Notifications	SMTP_HOST, SMTP_PORT, SMTP_USER, SMTP_PASSWORD, ALERT_FROM; enable only for authorized recipients.

Meta research access does not become usable by setting a generic token. Implement its adapter only after permitted execution/export rules are established. TikTok and Telegram credentials are not assumed. Document scopes, renewal, revocation and quota checks for each activated service.

Validate configuration at startup; redact secret values in diagnostics. Never request that real keys be pasted into the PDF, committed to Git or placed in NEXT_PUBLIC variables. Provide explicit “configuration required” states and a separate synthetic demo mode.

33 / CODEX / COPY THIS BRIEF

Codex master build instruction

Use this section as the initial coding brief with the complete PDF attached. Build PNSI, a production-oriented internal Tunisian health social-listening and infodemic-management application. Implement a dedicated Next.js/React/TypeScript frontend and FastAPI backend, PostgreSQL, private object storage and asynchronous workers. Do not use Streamlit as the product interface.

First inspect the existing repository and AGENTS.md. Preserve unrelated work, use an isolated branch when appropriate, and document what already exists. Treat this PDF as the product specification. Implement in vertical slices and complete meaningful tests before broadening scope. Use maintained compatible packages, lockfiles, migrations and typed API clients.

Implement all page families in sections 17-19, the case workflow in section 18, roles in section 3 and approval invariants in sections 20 and 35. All page controls must connect to real persisted backend behavior. Provide French, English, standard Arabic and reviewed Derja localization infrastructure; support Arabic RTL, Arabic-script Derja, Latin-script Arabizi and mixed content. If translations are provisional, mark them for review; do not claim full localization until checked.

Start with authorized imports and a permitted RSS connector plus a reviewed evidence corpus. Add optional connector interfaces, honest disconnected states and .env.example. Never invent API keys, access permissions, data, citations, official endorsements or successful deployments. Seeded data is visibly synthetic and isolated from live data.

Implement claim extraction, evidence retrieval, reviewer assessment, grounded multilingual drafts, exact-version approval, manually recorded publication, follow-up and reporting. Preserve uncertain/conflicting outcomes. Use provider adapters and local-processing defaults; never use model confidence as a truth score. External AI and public publishing stay disabled unless explicitly authorized.

Enforce backend object authorization, source approvals, document versions, redaction, source-health reporting and deletion propagation. Audit all material changes. Add tested loading/empty/error/partial states. Build the navy/teal/white institutional design from section 16, with keyboard accessibility and RTL verification.

For each delivery: report implemented requirements, commands run and results, scientific/translation review still needed, blocked connectors and exact configuration steps. Do not claim production readiness from a UI demo. Stop only for a genuine missing decision or access dependency while continuing independent authorized work. Finish with a reproducible local build, tests, screenshots, docs and a candid readiness report.

34 / CODEX / IMPLEMENTATION PLAN

Codex tasks, repository layout and deliverables

Build order	Deliverable and acceptance
1. Repository foundation	apps/web, apps/api, workers, shared schemas, infra/compose, docs, tests; lockfiles, lint/typecheck and safe .env.example.
2. Identity and data	Migrations for institutions, membership, source, content, claim, evidence and versioned review/action records; scoped API tests.
3. Complete case slice	Import -> claim -> evidence -> reviewer decision -> draft -> approval -> action; persists across reload and worker restart.
4. Language/UI system	Locale dictionaries, Arabic RTL, Derja/Arabizi handling, original/translation views and tested navigation.
5. Collection and evidence	RSS/import baseline, registry approvals, source-health states, document extraction and traceable retrieval.
6. AI adapters	Validated structured output; grounding; abstention; local and optional remote provider boundaries.
7. Decision tools	Narratives, priority explanation, alerts, response studio and separate approval queue.
8. Follow-up/reporting	Action schedule, comparable metrics, PDF/CSV exports and frozen snapshots.
9. Hardening and handover	End-to-end/access tests, backup restore, release evaluation, ministry demo and operator SOPs.

Suggested docs: README.md for startup; docs/architecture.md; docs/api-contract.md; docs/source-registry.md; docs/evidence-policy.md; docs/localization.md; docs/security.md; docs/deployment.md; docs/evaluation.md; docs/user-sops.md; docs/decisions.md; docs/readiness.md. Keep prompt templates versioned under an AI module; keep migration files in source control.

CI should run formatting/lint, frontend type checks, backend unit/integration tests, migration checks, build and critical browser workflows. Provide a deterministic synthetic seed with at least supported, contradicted, misleading, insufficient and conflicting cases; no mock medical verdict is represented as a real assessment.

35 / CONTRACTS

Machine-readable contracts and workflow invariants

Translate these field contracts into typed backend models and a generated frontend client. UUIDs are server generated; enum values are stable; timestamps use ISO 8601 UTC; text length and upload limits are explicit. Store unknown values as null with a missingness reason, not a guessed value.

Object	Required contract
Claim	id, institution_id, original_text, neutral_text, language_spans[], topic_ids[], claim_type, case_id, created_at, version.
Assessment	id, claim_id, proposed_status, review_status, evidence_links[], rationale, limitations[], model_run_id, reviewer_id/null, reviewed_at/null, version.
Evidence link	passage_id, document_version_id, relation=supports|contradicts|context, relevance_reason, applicability, source_approval_id.
Response variant	id, case_id, assessment_version_id, locale, script=arabic|latin, channel, body, citation_ids[], state, version.
Approval	id, response_variant_id, exact_version, assessment_version_id, evidence_snapshot_hash, approver_id, role, timestamp.
Action	id, approved_variant_version, institution_id, channel, performed_at, actor_id, permitted_reference/null, follow_up_plan.
Priority	value/null, harm, growth, engagement, exposure_proxy, component_missingness, completeness, policy_version, rationale.

Invariant 1: an approval cannot refer to a draft or superseded assessment. **Invariant 2:** factual edits create a new response version and invalidate its approval. **Invariant 3:** evidence IDs must resolve to approved document versions. **Invariant 4:** an action references an approved exact version, never only a mutable response ID.

Invariant 5: source deletion and institution authorization apply to search/index/export paths. **Invariant 6:** uncertain assessments cannot generate a definitive falsehood correction. **Invariant 7:** missing keys or failed jobs cannot silently substitute synthetic data. **Invariant 8:** follow-up scheduling and retry operations are idempotent. Enforce these rules transactionally in domain services and test direct API calls.

36 / ACCEPTANCE

Definition of done and acceptance checklist

Release area	Evidence needed to call it complete
Product	All required routes implemented; every control works; no placeholder action presented as real.
Data	At least one permitted real connector and authorized imports; honest coverage/freshness; persistence and idempotency.
Verification	Approved corpus; traceable passages; contradictory/absent evidence handled; scientific reviewer workflow.
Language	French/English/Arabic/Derja UI reviewed; Arabizi and mixed-content processing evaluated; RTL pages and exports checked.
Responses	Grounded drafts; separate language approval; exact-version enforcement and manual action recording.
Oversight	Scoped institution access, audit, role tests, data-handling decision and deletion process.
Follow-up	48h/7d/30d schedule; valid denominators; comparable coverage; no unsupported causal claim.
Operations	Reproducible deployment, secrets isolation, health checks, monitoring, tested restore and documented rollback.
Evaluation	Frozen benchmark, language/task metrics, errors, sample sizes and approved release thresholds.
Handover	SOPs, training, source/evidence owners, configuration instructions, readiness and limitation report.

Distinguish three statuses: **demonstration-ready** = polished labeled synthetic end-to-end experience; **pilot-ready** = authorized live data, reviewed corpus, evaluated models, working workflows and trained users; **production-ready** = institutional sign-off, funded operations, approved hosting/security/privacy and tested recovery.

Before ministry presentation, verify that screenshots and live demo match the actual build. Prepare a five-minute complete case walkthrough, a multilingual/RTL demonstration and a source-outage scenario. Explain what works, what awaits keys/permissions and what still needs scientific evaluation. Neither this specification nor a generated codebase can guarantee perfect medical accuracy.

37 / SCENARIO

Worked example: one claim from detection to follow-up

Illustrative scenario only. An analyst imports several permitted public messages claiming “Product Z prevents illness Y”. Product Z and illness Y are synthetic placeholders; no medical conclusion is asserted here.

Step	What the platform does
Collect	Records source, date, import rights and redacted text; marks manually imported coverage.
Understand	Separates “Does Z work?” from “Z definitely prevents Y”; extracts the prevention claim without losing negation.
Group	Links equivalent assertions across French, Arabic and Arabizi; preserves stance and original-language mentions.
Triage	Reviewer assesses potential harm; growth/engagement may be unknown, so priority completeness is visible.
Retrieve	Searches approved sources for Z/Y and prevention context; shows documents, dates and relevant passages.
Assess	No adequate document -> insufficient evidence. Relevant contradiction -> proposed contradicted status pending expert confirmation.
Draft	Creates language variants from the reviewed outcome. If unclear, explains uncertainty rather than inventing a correction.
Approve	Scientific reviewer checks citations; language reviewer checks meaning; institutional approver signs exact versions.
Act	Authorized person posts externally and records channel/time/version; no automatic publication.
Observe	At 48h/7d/30d, reports collected conversation change and coverage limits; no causal or population claim.

Expected UI behavior: the same case contains all original mentions, decisions and versions. A revised translation returns to review. A withdrawn evidence document reopens the assessment. A connector outage leaves the case accessible but marks new trend data partial. The example can be implemented as a deterministic test/demo fixture.

38 / REFERENCES

Research references I: evidence and public health

Access checked 11 October 2026 unless stated otherwise. Source links support specific facts; proposed system choices, weights and targets are the author's design recommendations. The source concept note is the user-supplied PNSI document, sections 1-11.

[R1] WHO social-listening ethics guidance

WHO: Social listening in infodemic management for public health emergencies - guidance on ethical considerations. Used as the ethical framing for public-health listening; not a claim of local legal compliance.

[R2] ONMNE publication entry point

ONMNE official site lists publications, thematic dossiers and institutional information. The homepage includes dated material; individually verify current documents and exclude unrelated or suspect content.

[R10] DPM role and mission

Direction de la Pharmacie et du Médicament: rôle et mission. Search surfaced the institutional mission page. Homepage fetch failed. Confirm current regulatory issuer, publication channels and documents directly before activating the source.

[R11] WHO guideline catalog

WHO guidelines is a starting point for selecting topic-relevant current recommendations, not a blanket domain truth guarantee.

[R12] NCBI discovery and usage guidance

A General Introduction to the E-utilities. NCBI documents retrieval and usage limits; without a key, the default is 3 requests/second, with a key generally 10. Discover records, then appraise documents and respect full-text rights.

Candidate local sources awaiting confirmation: Ministry publication portal (santetunisie.rns.tn), INEAS (ineas.tn) and INSP publication channels. Failed retrieval is documented, not interpreted as proof that the institution or source does not exist. No public API is asserted for these institutions.

39 / REFERENCES

Research references II: APIs and engineering

API policies and prices can change. Recheck official documentation and account-specific terms before enabling collection. Meta SRE execution and export limitations must be reviewed separately from access pricing.

Ref	Official source and supported point
R3	developers.google.com/youtube/v3/getting-started - official quota-limited Data API. Current English page separates 100 search calls/day from 10,000 units/day for other endpoints; verify project quota.
R4	docs.bsky.app/docs/advanced-guides/api-directory - public Bluesky AppView endpoints; this does not establish Tunisian relevance.
R5	docs.joinmastodon.org/methods/timelines - public timeline API; instance authentication/policy can restrict access.
R6	docs.x.com/x-api/getting-started/pricing - pay-per-use; post reads currently $0.005 each, subject to change.
R7	support.reddithelp.com/hc/en-us/articles/42728983564564-Responsible-Builder-Policy - explicit API approval and RFR research access; updated 5 October 2026.
R8	developers.facebook.com/docs/content-library-and-api/get-access - no access/compute fees for interface and Meta SRE API; eligibility/approval required.
R9	transparency.meta.com/researchtools/meta-content-library - research archive access; not a general open social-data API.
R13	nextjs.org/docs/app/guides/internationalization - routing/rendering of localized content.
R14	fastapi.tiangolo.com/deployment/docker - container deployment guidance.

These references justify access constraints and the available engineering foundations. They do not establish that this specific project is eligible, licensed, operational or performance-validated. Website/RSS reuse, Telegram and TikTok integration remain source-specific research tasks.

40 / SOURCE ALIGNMENT

Traceability to the supplied concept note

Concept note requirement	Implementation section / treatment
National social listening and six layers (3-5)	Sections 4, 6-7, 9-11, 13 and 17-22 translate the full operational chain.
Facebook/TikTok/Instagram/X/etc. (5.2)	Section 6 access matrix; gated connectors rather than promised unrestricted coverage.
Languages and Arabizi (5.2-5.3)	Section 12; Italian planned extension; user-required Derja included.
OCR/ASR (5.3)	Sections 9 and 25; optional permitted media processing, separately evaluated.
Rumors, narratives and emotions (5.4)	Sections 10-11 and 25-26; model proposals distinct from expert verdicts.
Actor maps + anonymity (5.2/5.4/6)	Section 23 resolves conflict through aggregate baseline and a documented decision.
Temperature index, dashboards/reports (5.5)	Sections 11 and 16-22; experimental index, visible components and coverage.
Alerts and reviewed responses (5.6)	Sections 11, 18-20, 27 and 35; exact-version approval and abstention.
48h/7d/30d actions and impact (5.7)	Sections 21-22 and 37; trends distinguished from causal effects.
Multi-institution / local hosting (6)	Sections 3, 13-15, 23-24 and 29.
12-month phasing, training/staffing (7-9)	Sections 28-31 and 34; training/SOP ownership and timeline arithmetic addressed.
Future epidemiological integration (10)	Section 21; authorized aggregate schema only, no assumed national-system API.

Critical additions beyond the note: curated document-level evidence governance, claim verification/abstention, access feasibility, honest metric denominators, language evaluation, transactional approval invariants, code contracts, operational tests and implementation instructions. These additions make the concept actionable without claiming that unresolved permissions or scientific validation are already complete.

41 / HANDOVER

Glossary and handover package

Term	Practical meaning
Infodemic	Overabundance of information, including accurate and inaccurate content, that complicates health understanding and action.
Social listening	Systematic analysis of available conversations, questions and concerns within explicitly defined sources.
Claim / narrative	A checkable assertion / a broader repeated story or theme containing one or more claims.
Rumor / misinformation	An unverified circulating claim / false or misleading information; rumor is not itself a false verdict.
Evidence whitelist	Approved source/document registry with scope, currentness, permissions and scientific review.
RAG	Retrieval-augmented generation: evidence retrieval followed by constrained model output; requires evaluation and review.
Derja / Arabizi	Tunisian dialect / Latin-letter-and-number writing conventions, often mixed with French or Arabic.
RTL	Right-to-left layout for Arabic-script content; not the direction of Latin-script Arabizi.
Provenance	Traceable origin, version, extraction and decision history of a record.
Abstention	Explicitly declining a verdict when evidence or model reliability is insufficient.
Coverage	What sources, periods, queries and languages were actually collected; not the whole public conversation.

What must accompany the implemented platform

A working repository, dependency lockfiles, migrations, safe configuration examples, tested deployment instructions, source registry, reviewed evidence corpus, localization glossary, dataset/evaluation cards, access/retention policy, reviewer SOPs, training materials, backup/incident runbooks and a readiness report.

This PDF is the master planning and coding reference. It contains no real API credentials, real clinical verdicts, deployment guarantees or unapproved institutional branding. The next implementation step is to give Codex this document and the repository, begin with section 33, and deliver the tested vertical slice before expanding to gated data sources.

42 / APPENDIX / TAXONOMY

Appendix A: taxonomy and multilingual query design

Keep the full health scope, while limiting the first monitored themes according to partner priorities and evidence availability. Each topic has a stable ID, parent category, multilingual names, synonyms, excluded meanings, approved seed queries, responsible reviewer and version.

Topic family	Examples of subtopics
Prevention and vaccination	Vaccine safety, schedules, eligibility, prevention measures and community questions.
Infectious disease	Respiratory/vector-borne illness, outbreaks, symptoms, transmission and testing.
Chronic conditions	Diabetes, cardiovascular conditions, cancer, kidney disease and long-term care.
Medicines and treatments	Safety alerts, availability claims, dosing claims, counterfeit concerns and alleged cures.
Mental/reproductive health	Stigma, access, prevention, maternal/child health and information gaps.
Nutrition and environment	Supplements, food safety, heat, pollution and environmental health concerns.
Services and trust	Care access, institutional communication, complaints, questions and legitimate criticism.

Query construction combines a health concept with optional local context, spelling/script variants and narrative terms. A French seed such as “vaccin effets secondaires Tunisie” is only an example of a search, not a medical claim. Tunisian clinicians and language reviewers must supply Arabic/Derja/Arabizi equivalents and disambiguation rules; do not assume one machine translation captures all variants.

Maintain a reviewed term dictionary including common transliterations, product generic/brand names where relevant, code-switch phrases and negative keyword rules. Test each query on labeled samples for relevance and missed concerns. Do not require the word “Tunisia” in every post: local publisher context can establish relevance, while dialect alone cannot establish residence.

Version query changes and annotate affected charts. Source expansion and keyword changes can increase counts without a true conversation surge. Preserve query ID and version on each collection run.

43 / APPENDIX / MULTIMEDIA

Appendix B: permitted image and audio processing

Audiovisual capability is part of the national ambition, but API metadata access does not automatically grant permission to download videos or audio. Enable processing only for authorized media from allowed sources or institutional uploads. Prioritize relevant items rather than processing every file.

Processing steps

Validate type/size and scan; place in private temporary storage; extract permitted audio; detect language segments; transcribe speech with timestamps and uncertainty; sample relevant frames; run OCR on text regions; combine aligned text without inventing unseen content; redact identifiers; link derived claims to exact time/frame/page; delete temporary media according to rights and retention.

Separate component evaluation

Compare transcription and OCR candidates on Tunisian speech, noise, accents, code-switching and low-resolution overlays. Measure word/character errors plus medically important omissions: negation, numbers, units, product names and the speaker’s stance. Good overall transcription accuracy can still miss a critical “not”.

Let analysts correct transcripts and inspect permitted time segments. Store the original derived transcript, corrected version, extraction model/version and confidence limits. Low-quality material is queued for manual review; missing audio or unreadable text yields unavailable/uncertain rather than an invented transcript.

Keep OCR/ASR as optional asynchronous worker modules with independent resource budgets. A text-first pilot is fully usable without them. Do not label a video verified solely from its title, thumbnail or comments; make the scope of the actual reviewed material explicit.

44 / APPENDIX / REGISTRY

Appendix C: evidence registry starter template

Codex should implement these as database-backed forms and exportable structured records, not a hard-coded domain list. The seed remains pending until a named scientific reviewer approves the exact scope and documents.

Field	Example starter value
registry_id / organization	EVID-ONMNE-001 / ONMNE
canonical_host / allowed_paths	www.onmne.tn / explicit reviewed publication paths to be supplied
purpose / jurisdiction	Local surveillance and public-health guidance / Tunisia
allowed_document_types	Official circulars, thematic guidance, dated institutional publications
topics / languages	Partner-selected pilot topics / language detected per document
authority_scope	Only statements within the issuer’s remit; no blanket medical authority
approval_state / approver	pending / assigned institution scientific reviewer
rights / collection_method	to confirm / authorized upload or permitted page retrieval
review_interval / expiry	Proposed 90-day source review; urgent documents checked more often
document_id / version	Generated per approved document; immutable checksum and dates
withdrawal_policy	Suspend outdated material; mark affected assessments for review

Document approval form asks: Is the issuer authentic? Is this an official document? Is it in scope? Is the edition current? Is the full relevant passage available? Are reuse/storage allowed? Does the population and jurisdiction fit? Are limitations or contradictions recorded? If unresolved, keep pending.

Bootstrap workflow: request documents from partners -> register candidates -> verify authenticity and rights -> scientific scope review -> document ingestion -> sampled extraction QA -> index -> test retrieval on reviewed claims -> activate for proposals. A WHO or ONMNE name in a URL or document title alone is insufficient.

45 / APPENDIX / SOP

Appendix D: daily operating procedures

Procedure	Owner and minimum steps
Morning source check	Operator checks last success, outages, quotas and deletion jobs; analyst sees coverage changes before interpreting trends.
Triage	Analyst reviews new cases/questions, validates groups, assesses potential harm, assigns owners and due times.
Verification	Scientific reviewer inspects claim context and passages, considers contradictory evidence, confirms status or requests investigation.
Draft review	Communication/language reviewers check evidence alignment, tone, translation equivalence and approved channel guidance.
Institutional approval	Authorized approver accepts exact version or requests changes; edits restart affected checks.
Publication recording	Responsible person records actual channel/time/version; system creates follow-up tasks once.
Urgent escalation	High potential harm is routed to a designated institution contact; platform notifications require configured recipients.
Evidence maintenance	Source steward checks urgent updates and supersession; impacted assessments are reopened.
Weekly learning	Team reviews mistakes, unresolved cases, language errors, coverage gaps and action observations.
Monthly oversight	Review release quality, operating cost, source permissions, retention, training and policy changes.

Training uses synthetic complete cases before live access: one contradicted claim, one unresolved claim, one question, one legitimate complaint and one mixed-language example. Reviewers practice citation checks, approval invalidation, correction after publication and interpreting source outages.

Handover must name actual owners, backups and escalation windows. Do not infer 24/7 staffing, notify people automatically or invent institutional recipients. A national service requires sustained scientific and operational capacity in addition to the software.